

Pin Documentation

CommandBrick - ESP32-S3 Hardware Connections

1. Introduction

This document describes all electrical connections used in the **CommandBrick** project.

The ESP32-S3 serves as the central controller and communicates with all connected peripherals using GPIO, I²C, and power connections. This document provides an overview of the implemented hardware modules, their electrical interfaces, and the pin assignments required to assemble and operate the device.

2. Implemented Hardware Modules

The CommandBrick prototype is built using the following hardware components.

Component	Model	Description
Microcontroller	ESP32-S3 Development Board	Main controller responsible for processing sensor input and Bluetooth HID communication.
Capacitive Touch Controller	MPR121	12-channel capacitive touch controller used to detect touches on the capacitive electrodes.
Touchpad	Cirque Pinnacle TM035035 Capacitive Touchpad	Provides finger position and movement data for Bluetooth mouse control.
Bluetooth Interface	Integrated ESP32-S3 BLE	Provides Bluetooth Low Energy communication and Bluetooth HID functionality.
Haptic Feedback	Vibration Motor	Generates tactile feedback whenever a capacitive button is activated.
Battery	3.7 V Li-Ion Battery	Rechargeable power source for the complete device.
Charging Module	TP4056 with Protection	Charges the Li-Ion battery and provides protection against overcharging, over-discharging, and short circuits.

3. System Overview

The Command Brick hardware consists of the following main components:

- ESP32-S3 Development Board
- Cirque Pinnacle TM035035 Capacitive Touchpad
- MPR121 Capacitive Touch Controller
- Vibration Motor
- 3.7 V Li-Ion Battery
- TP4056 Charging & Protection Module

4. ESP32-S3 Pin Assignment

ESP32-S3 Pin	Connected Device	Function
GPIO 8	MPR121 SDA / Touchpad SDA	I ² C Data
GPIO 9	MPR121 SCL / Touchpad SCL	I ² C Clock
GPIO 10	Cirque Pinnacle TM035035 DR	Data Ready Interrupt
GPIO 5	Vibration Motor	Haptic Feedback Output
3.3 V	MPR121	Power Supply
3.3 V	Cirque Pinnacle TM035035	Power Supply
GND	All Components	Common Ground

5. I²C Communication Bus

The CommandBrick uses a shared I²C bus to communicate with the touch controller and the touchpad.

Device	I ² C Address
MPR121	0x5A
Cirque Pinnacle TM035035	0x2A

The ESP32-S3 operates as the I²C master and continuously polls both devices during operation.

6. Cirque Pinnacle TM035035 Connections

Touchpad Pin	ESP32-S3 Pin
SDA	GPIO 8
SCL	GPIO 9
DR	GPIO 10
VCC	3.3 V
GND	GND

Purpose

The Cirque Pinnacle TM035035 capacitive touchpad is responsible for mouse cursor control.

It continuously detects finger movement and position. The ESP32 reads this information through the I²C interface, processes the movement, and sends Bluetooth HID mouse reports to the connected computer.

7. MPR121 Connections

MPR121 Pin	ESP32-S3 Pin
SDA	GPIO 8
SCL	GPIO 9
VCC	3.3 V
GND	GND

Purpose

The MPR121 monitors twelve capacitive electrodes connected to the device.

Whenever a user touches an electrode, the controller reports the event to the ESP32, which converts it into the corresponding keyboard or mouse command.

8. Capacitive Pin Assignment

Pin	Assigned Function
P0	Left Mouse Click
P1	Right Mouse Click
P2	Space
P3	Enter
P4	Tab
P5	Backspace
P6	Ctrl
P7	Alt
P8	Shift
P9	Arrow Up
P10	Arrow Down
P11	Open Windows On-Screen Keyboard (Ctrl + GUI + O)

9. Vibration Motor

Motor Pin	Connection
Positive	GPIO 5
Negative	GND

Purpose

The vibration motor provides tactile feedback to the user.

Whenever a capacitive button is successfully detected, the ESP32 briefly activates the motor to confirm that the input has been registered.

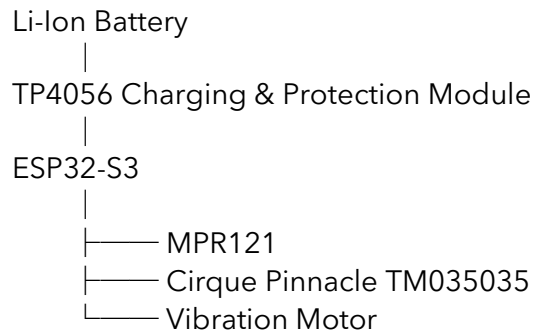
10. Power Supply

The CommandBrick is powered by a rechargeable **3.7 V Li-Ion battery**.

The battery is connected to a **TP4056 charging and protection module**, which safely charges the battery while protecting it against overcharging, over-discharging, and short circuits.

The ESP32-S3 receives power from the TP4056 module and distributes it to all connected peripherals.

Power Distribution



11. Communication Summary

Interface	Connected Devices
I ² C	MPR121, Cirque Pinnacle TM035035
GPIO	Vibration Motor
Bluetooth Low Energy (BLE)	Computer, Tablet, Smartphone
Power	Li-Ion Battery and TP4056 Module

12. Notes

- All components share a common **GND** connection.
- The ESP32-S3, MPR121, and Cirque Pinnacle TM035035 operate at **3.3 V logic levels**.
- The MPR121 and the Cirque Pinnacle TM035035 share the same I²C communication bus.
- The **Data Ready (DR)** pin of the touchpad is connected to **GPIO 10**, allowing the ESP32 to detect when new touch data is available.
- Bluetooth communication is handled entirely by the ESP32-S3 using the **NimBLE HID** library, allowing the device to function as a standard Bluetooth keyboard and mouse without requiring additional drivers on the host computer.